

Write the output of the following code snippets.

<pre>class MyArray { private: int* arr; int size; public: MyArray(int s){ size = s; arr = new int[size]; for (int i = 0; i < size; i++) { arr[i] = i + 1; } cout << "In constructor" << endl; } MyArray(const MyArray& other){ size = other.size; arr = new int[size]; for (int i = 0; i < size; i++) { arr[i] = other.arr[i]; } cout << "In copy constructor" << endl; } void display() const { for (int i = 0; i < size; i++) { cout << arr[i] << " "; } cout << endl; } void setVal(int index, int val) { arr[index] = val; } ~MyArray() { delete[] arr; } }; int main() { MyArray a(6); MyArray b(a); a.setVal(2, 0); b.setVal(3, 0); a.display(); b.display(); }</pre>	In constructo r. In copy constructo r. 120456 123056
---	---

<pre> int someSearchFunction(int arr[], int size, int target, int& check) { check = 0; for (int i = size - 1; i >= 0; i--) { check++; if (arr[i] == target) return i; } return -1; } int main() { int arr[] = { 3, 7, 1, 9, 5, 2, 0, 6, 8 }; int size = 9; int target = 0; int check = 0; int result = someSearchFunction(arr, size, target, check); cout << result << endl; cout << check << endl; return 0; } </pre>	6 3
<pre> int binarySearch(int arr[], int size, int target, int& compCount) { int low = 0, high = size - 1; compCount = 0; while (low <= high) { int mid = (high + low) / 2; cout << mid << endl; compCount++; if (arr[mid] == target) return mid; else if (arr[mid] < target) low = mid + 1; else high = mid - 1; } return -1; } int main() { int arr[] = { 1, 3, 5, 7, 9, 11, 13, 15 }; int size = 8; int target = 3; int compCount = 0; int result = binarySearch(arr, size, target, compCount); cout << result << endl; cout << compCount << endl; return 0; } </pre>	3 1 1 2

<pre> #include <iostream> using namespace std; void bubbleSort(int arr[], int size) { for (int i = 0; i < size - 1; i++) { cout << "Outer loop" << endl; bool swapped = false; for (int j = 0; j < size - i - 1; j++) { if (arr[j] > arr[j + 1]) { int temp = arr[j]; arr[j] = arr[j + 1]; arr[j + 1] = temp; swapped = true; } } if (!swapped) { break; } } } int main() { int arr[] = { 1, 3, 8, 12, 22, 56, 90 }; int size = 7; bubbleSort(arr, size); return 0; } </pre>	Outer loop
<pre> #include <iostream> using namespace std; void swapNums(int &x, int &y) { int z = x; x = y; y = z; } int main() { int firstNum = 10; int secondNum = 20; cout << "Before swap: " << "\n"; cout << firstNum << secondNum << "\n"; swapNums(firstNum, secondNum); cout << "After swap: " << "\n"; cout << firstNum << secondNum << "\n"; return 0; } </pre>	Before swap: 1020 After swap: 2010

<pre> bool check(const char a[], const char b[]) { int i = 0; while (a[i] != '\0' b[i] != '\0') { if (a[i] != b[i]) return false; i++; } return true; } int main() { char str1[] = "Hello, World!"; char str2[] = "Hello, C++!"; if (check(str1, str2)) cout << "Okay" << endl; else cout << "Not okay" << endl; return 0; } </pre>	Not okay
<pre> int main() { int arr[] = { 1, 2, 3, 4, 5 }; int size = 5; int something = 0; for (int i = 0; i < size; i++) { something += arr[i]; } cout << something << endl; return 0; } </pre>	15
<pre> #include <iostream> #include <vector> using namespace std; int main() { vector<int> vec = { 10, 20, 30, 40, 50 }; int something = 0; cout << "Elements in vec:" << endl; for (auto element : vec) { something += element; } cout << something; return 0; } </pre>	150

<pre> #include <iostream> using namespace std; int main() { int arr[] = { 5, 7, 2, 9, 1, 12 }; int size = 6; int x = arr[0]; int xi = 0; for (int i = 1; i < size; i++) { if (arr[i] > x) { x = arr[i]; xi = i; } } cout << x << endl; return 0; } </pre>	12
<pre> #include <iostream> using namespace std; int main() { int arr[] = { 5, 7, 2, 9, 1, 12 }; int size = 6; int x = arr[0]; int xi = 0; for (int i = 1; i < size; i++) { if (arr[i] > x) { x = arr[i]; xi = i; } } cout << xi << endl; return 0; } </pre>	5